

Professional Summary

Machine Learning Engineer with experience building production-grade LLM platforms, agentic AI systems, and scalable inference infrastructure across Kubernetes, vLLM, Ray, and cloud environments. Strong focus on turning emerging AI capabilities into reliable internal platforms, cost-efficient model-serving systems, and developer workflows that improve engineering productivity.

Experience

Panasonic Avionics

Machine Learning Engineer - I

July 2025 – June 2026

Machine Learning Engineer - II

Promoted June 2026 – Present

- Developed an end to end routing solution for LLM's designed specifically for coding agents like Claude Code, Kilo Code, Cursor improving cost utilization by 4x.
- Initiated, Managed, and iterated on the organization's LLM inference platform, hosting a mixture of open-source and closed-source models, increasing productivity by 60%. The platform exposed a unified endpoint supporting a wide range of LLM SDKs.
- Built an agentic AI platform that standardized the development and deployment of production-grade AI agents, reducing agent creation time from days to under 10 minutes.
- Architected a Kubernetes controller platform for organization-wide LLM deployments, providing automated lifecycle management, budget controls, centralized observability, and operational controls across development and production environments, reducing debugging and incident resolution time by 90%.

Calsoft, Inc.

Data Scientist (Consultant for Panasonic Avionics)

January 2025 – July 2025

- Built input/output assessment guardrails using open and closed source technologies, achieving an 88% safer AI system.
- Engineered a WAF-fortified accelerated inference pipeline with vLLM, reaching 80 tok/s on dense models for proprietary coding and application workflows.

Softsensor.AI

India

Associate Data Scientist

September 2023 – January 2025

- **Cyber Security Chatbot:** Developed core logic for Text2SQL and unstructured MITRE document query workflows for precise log and attack insights.
- **Document Translation:** Created end-to-end translation software preserving tables and diagrams while translating diverse document formats.
- **Claims Automation:** Led full-cycle unstructured-to-structured transformation for letters, emails, and SRRs with interlinked document relationships, reducing extraction cost by up to 85
- **R&D:** Contributed to research in topic modeling, ColBERT reranking, LLM finetuning, and multimodal data ingestion pipelines.

Blackcoffer-Consulting

Remote

Associate Data Scientist

March 2022 – February 2023

- Developed a computer vision solution for satellite imagery analysis, enabling automated detection and localization of vehicles from high-resolution satellite images.
- Designed interactive data visualization dashboards in Google Data Studio for chemical and environmental datasets, enabling stakeholders at One.Five to identify pollution trends and make data-driven operational decisions.
- Engineered a Flask-based NLP application for document summarization and lexical simplification, transforming lengthy documents into prioritized bullet-point summaries that surfaced critical information for end users.

- Conducted exploratory data analysis and market intelligence on Airbnb datasets, uncovering insights into property types, regional demand patterns, and customer preferences across markets and zip codes.
- Built business intelligence dashboards for an e-commerce platform, consolidating operational and sales metrics into actionable visualizations that supported strategic and data-driven decision making.

Leadership, Projects & Activities

Panasonic Avionics

MLE-I

- Mentored an intern in AWS and AI-driven engineering practices, accelerating her ability to independently learn, troubleshoot complex issues, and adopt emerging technologies.

Personal Projects and Freelance

- Contributed to the open-source Lighteval & LlamaIndex ecosystem by implementing enhancements to documentation, data-ingestion workflows and advanced storage backends, including knowledge graph integrations, improving the framework's extensibility and data processing capabilities. (Contrib #1, Contrib #2, Contrib #3)
- Won a Generative AI Hackathon by developing an innovative system that transformed plain-text narratives into interactive visual game experiences using open-source generative AI models, showcasing end-to-end multimodal content generation. (Project)
- Developed an open-source agentic AI framework that enabled quantized LLMs to execute tool-calling workflows on CPU-only systems, leveraging multi-threaded inference to deliver low-cost, high-performance agent execution. (GitHub)
- Built an intelligent chatbot for the University of Central Missouri by constructing a custom NLP pipeline from institutional FAQs and social media data, incorporating sentiment-aware intent classification to deliver contextually relevant responses. (Freelance)

Education

Amity University

B.Tech, Computer Science

India

18th Jan 2024

Relevant Coursework:

- Neural Networks and Deep Learning Course by DeepLearningAI
- Azure Essentials Masterclass
- Coursera - Machine Learning Course Certificate
- Microsoft Azure: AI-900 Fundamentals

Jayshree Periwal High School - Grade XII

Specialization - PCM

India

2019

Skills & Interests

Programming Languages: Python, JavaScript

AI & Machine Learning: Large Language Models (LLMs), Agentic AI, RAG, Fine-Tuning, Prompt Engineering, NLP, Computer Vision, Topic Modeling, Knowledge Graphs, ColBERT, Text2SQL, Multimodal AI

AI Infrastructure & MLOps: Kubernetes, vLLM, Ray, Model Serving, Inference Optimization, Observability, AI Guardrails, WAF, Docker, CI/CD

Cloud & Data Engineering: AWS, Microsoft Azure, Data Pipelines, ETL, Data Visualization, Google Data Studio

Frameworks & Tools: LlamaIndex, LangChain, Flask, Git, Linux

Interests: Strength Training, Mixed Martial Arts (MMA), Chess